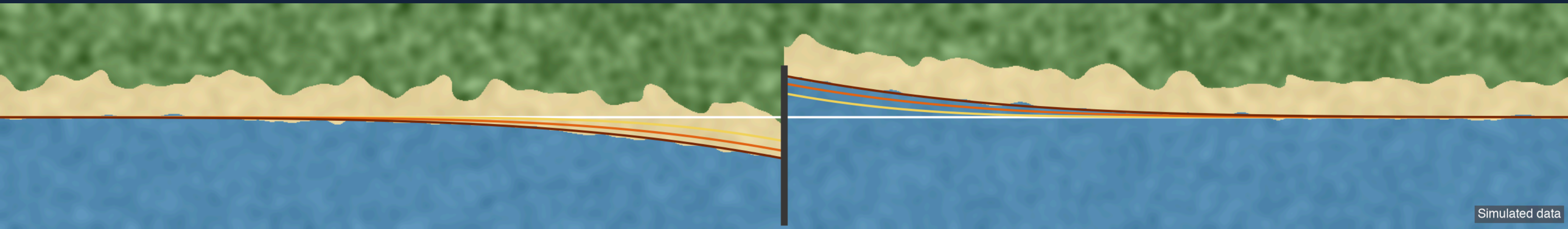
ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Coastal Erosion Monitoring with Remote Sensing

Tracking moving shorelines with forty years of satellite images

AKSHAY RAJ MANOCHA



GOAL OF THIS LECTURE

Use satellite images to find the shoreline, measure how fast it moves, and judge what that means for people and structures on the coast.

Erosion processes, extracting shorelines, rates of change and their uncertainty, harbours and groynes, and Indian case studies.

THIS LECTURE

Outline

01 Coastal erosion and why we monitor it
Processes, causes, India’s coast, and what satellites add

02 Mapping the shoreline from space
Shoreline indicators, satellites, water indices, tides and radar

03 Measuring shoreline change
Transects and rates, uncertainty, structures, and elevation change

04 Case studies
The world’s beaches, India’s coast, Puducherry and Satabhaya

05 Engineering practice
Uses, coastal management, a workflow, limits and exercises

01

PART 1

Coastal erosion and why we monitor it

Processes, causes, India’s coast, and what satellites add

COASTAL EROSION

A beach is a sand budget

A beach grows when more sand arrives than leaves, and erodes when more leaves. Erosion means the shoreline moves landward over years.

Sand in

Rivers, cliffs, and sand carried along the coast by waves.

Sand out

Longshore drift, storms dragging sand offshore, wind, sand mining.

Short-term vs long-term

Storms and monsoon seasons move the shoreline tens of metres and back. Engineers need the long-term trend underneath.

COASTAL EROSION

Why coasts erode

Waves and storms

Cyclones and monsoon waves remove sand in days.

Sea-level rise

About 3.7 mm a year globally (2006–2018, IPCC AR6).

Less river sand

Dams trap sediment; sand is mined from rivers and beaches.

Coastal structures

Harbours, groynes and seawalls cut off or reflect drift.

Sinking deltas

Subsidence lowers the land behind the coast (Lecture 6).

Loss of mangroves

Roots that trap sediment and absorb waves are removed.

COASTAL EROSION

India’s eroding coast

33.6%
Of the mainland coast eroded, 1990–2018

26.9%
Gained land (accreted)

39.6%
Stable

≈ 60%
Of West Bengal’s coast eroded: the worst state

Hotspots

Puducherry (about 56% eroding), Kerala and Tamil Nadu (about 41% each).

Measured from space

The National Centre for Coastal Research mapped the whole coast from satellite images, checked in the field.

Sources: NCCR (Ministry of Earth Sciences), National Assessment of Shoreline Changes; PIB; Down To Earth.

COASTAL EROSION

Why satellites for coasts

Since 1984
Landsat archive: 40 years of shorelines

10 m
Sentinel-2 pixels, every 5 days

Free
Landsat, Sentinel and the tools to process them

Whole coasts
Thousands of km at once

Long records
Field surveys cover a few beaches for a few years; satellites cover every beach for decades.

Before and after
Measure a harbour’s effect, a cyclone’s damage or a nourishment project’s success.

But not centimetres
Pixel size, tides and waves limit precision to metres: fine for trends over years.

02

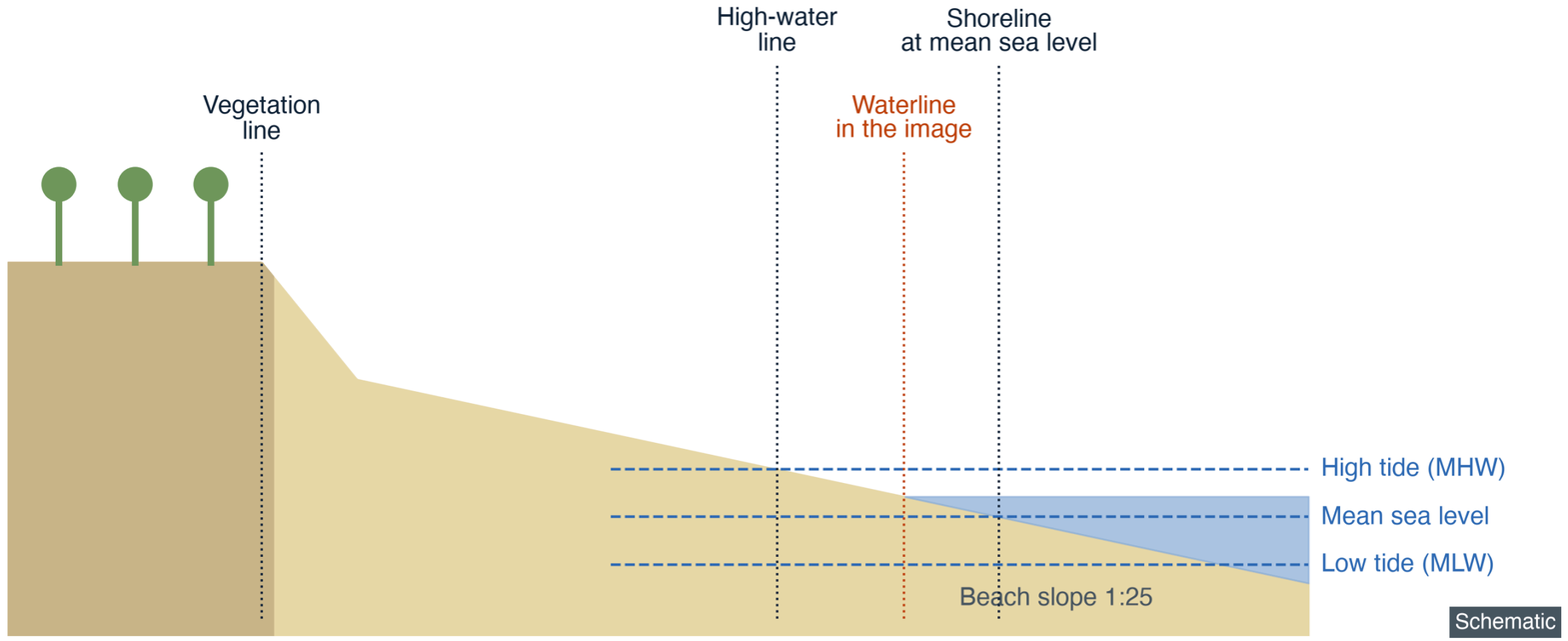
PART 2

Mapping the shoreline from space

Shoreline indicators, satellites, water indices, tides and radar

MAPPING

Which ‘shoreline’?



Schematic. Each indicator gives a different position; always say which one you mapped.

Waterline: where water meets sand when the image was taken; moves with the tide.

Datum shoreline: the waterline corrected to a fixed level, such as mean sea level.

Vegetation line: stable, but moves only after big storms.

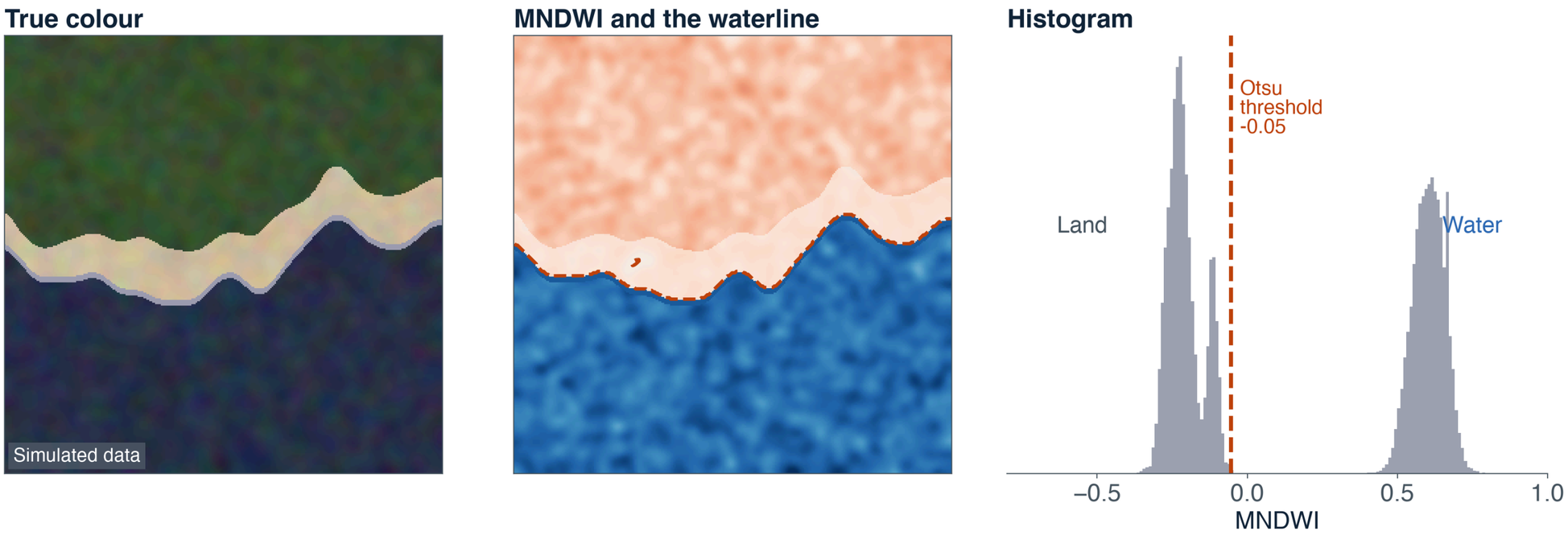
MAPPING

Data for shoreline mapping

Source	Detail	Notes
Landsat 5, 7, 8, 9	30 m	Since 1984: the long record
Sentinel-2	10 m	Since 2015; every 5 days
Sentinel-1 (SAR)	10 m	Through cloud; since 2014
Cartosat, Pléiades, WorldView	Under 1 m	Detail for a town or harbour; not free
Drones and lidar	Centimetres	Beach shape and volume; small areas
Old maps and air photos	Varies	Shorelines before satellites

MAPPING

Finding the water’s edge



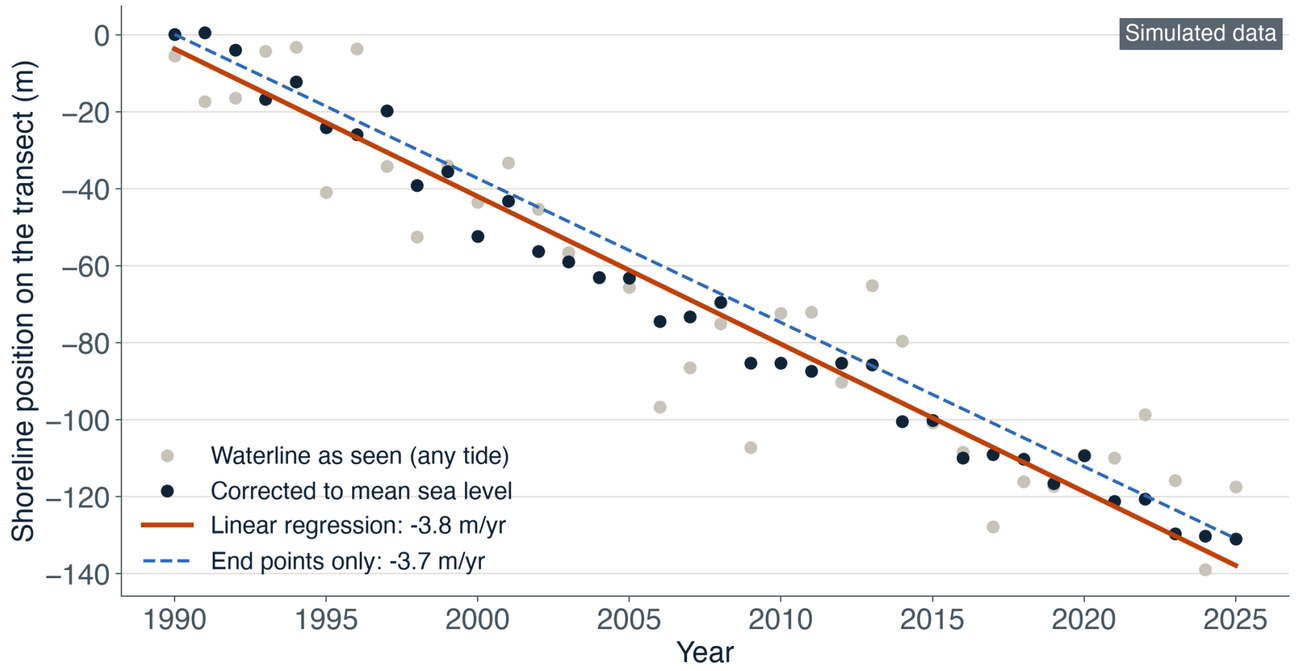
Simulated scene. The threshold between land and water peaks is chosen automatically (Otsu): here –0.05.

$$\mathrm{MNDWI} = (\mathrm{Green} - \mathrm{SWIR}) \div (\mathrm{Green} + \mathrm{SWIR})$$

Water absorbs short-wave infrared; sand reflects it. Sub-pixel methods place the edge to a fraction of a pixel.

MAPPING

Tides move the waterline



Simulated transect, one image a year at random tides.

Shift = tide height ÷ beach slope

A 1 m tide on a 1:25 beach moves the waterline 25 m: more than a Landsat pixel.

Correcting to mean sea level cuts the scatter here from about 14 m to 5 m.

Tide heights come from tide gauges or global tide models (FES).

MAPPING

Radar for the monsoon coast

Through cloud

The monsoon hides much of India’s coast from optical satellites for months; radar sees it every few days.

Calm water is dark

As in flood mapping, smooth water reflects radar away from the satellite (Lecture 7).

But waves are bright

Breaking waves and wind roughen the sea, blurring the edge. Use calm days and averages.

Best together

Radar for frequent, all-weather images; optical for the clearest edge.

03

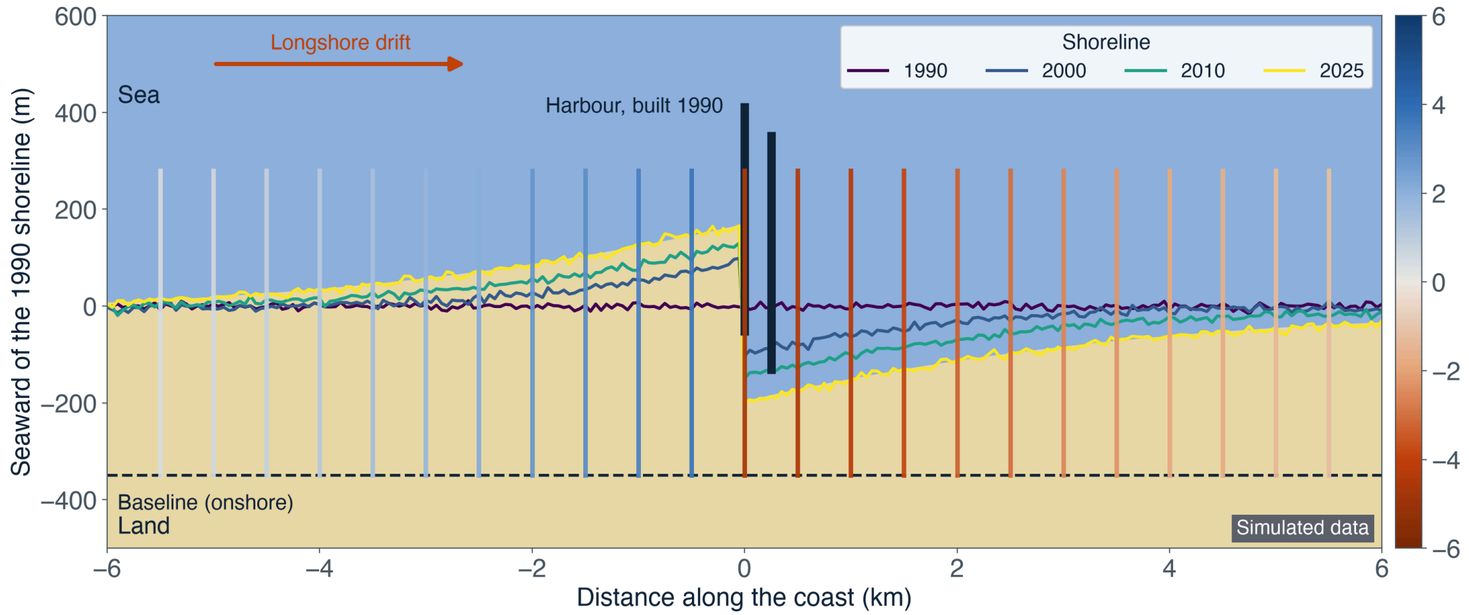
PART 3

Measuring shoreline change

Transects and rates, uncertainty, structures, and elevation change

CHANGE

Transects from a baseline



Simulated coast, 12 km, one shoreline a year (four shown).

Draw a baseline on land, then transects across the shore every 50–100 m.

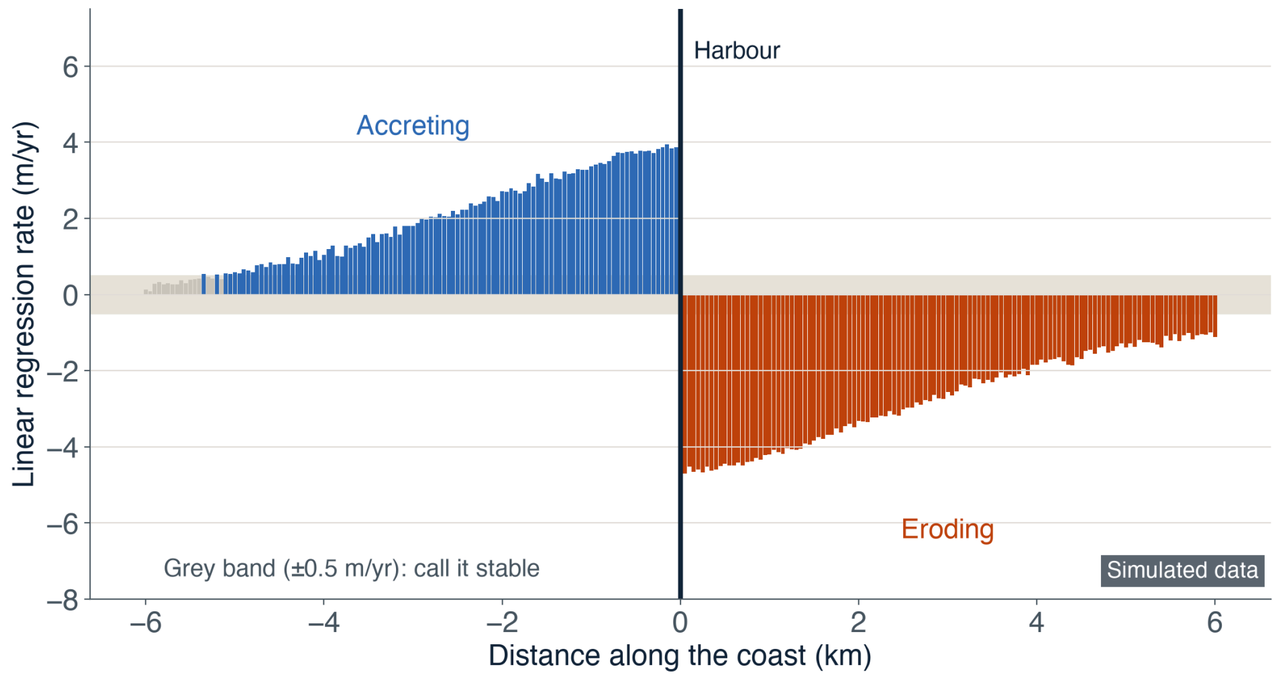
Where each shoreline crosses each transect gives a position for each year.

Fit a rate on every transect: the colour shows it.

The USGS Digital Shoreline Analysis System (DSAS) automates this in GIS.

CHANGE

Rates of shoreline change



Simulated: linear regression rate on each transect.

NSM: net shoreline movement, last minus first (m)

EPR: end-point rate = NSM ÷ years (m/yr)

LRR: slope of a straight line through all years (m/yr)

On the transect shown earlier: NSM −131 m; EPR −3.7 m/yr; LRR −3.8 m/yr.

LRR uses every year, so one odd image matters less.

CHANGE

How sure is a rate?

$$\sigma(\text{rate}) \approx \sqrt{(\sigma_1^2 + \sigma_2^2)} \div \text{years}$$

For an end-point rate, from the position errors σ_1 and σ_2 of the two shorelines.

± 15 m

Typical Landsat shoreline error

± 5 m

Typical Sentinel-2 shoreline error

Rule of thumb

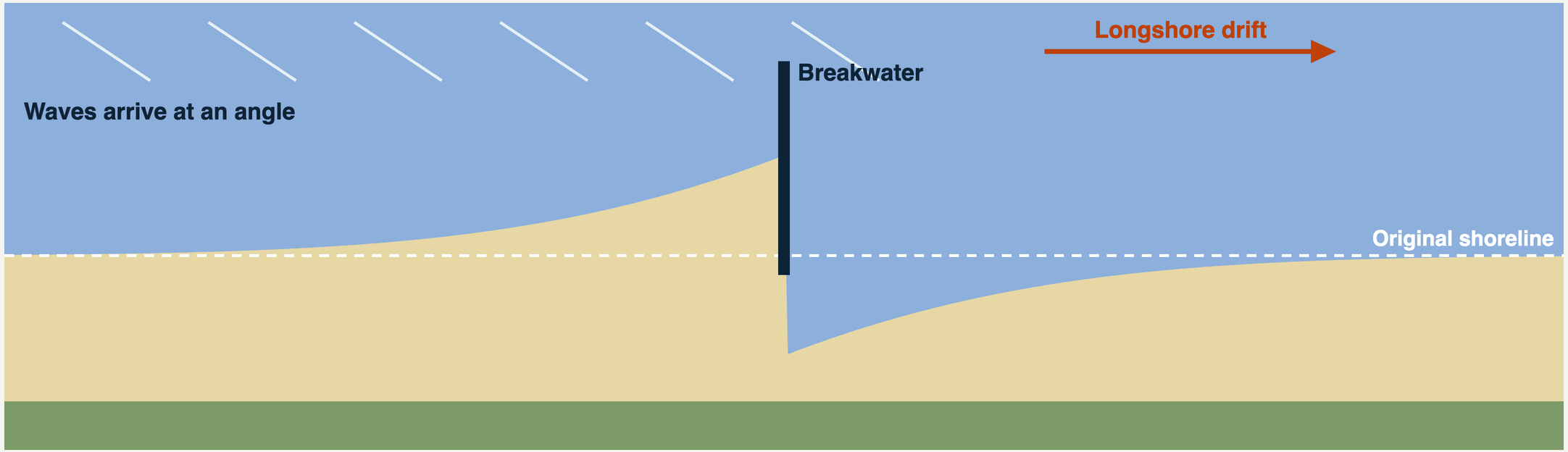
Rates smaller than about ±0.5 m/yr over a few decades are hard to separate from noise: call them ‘stable’.

Other errors

Tide and wave run-up, seasons, storms just before an image, and image misregistration.

CHANGE

What a harbour does to the coast

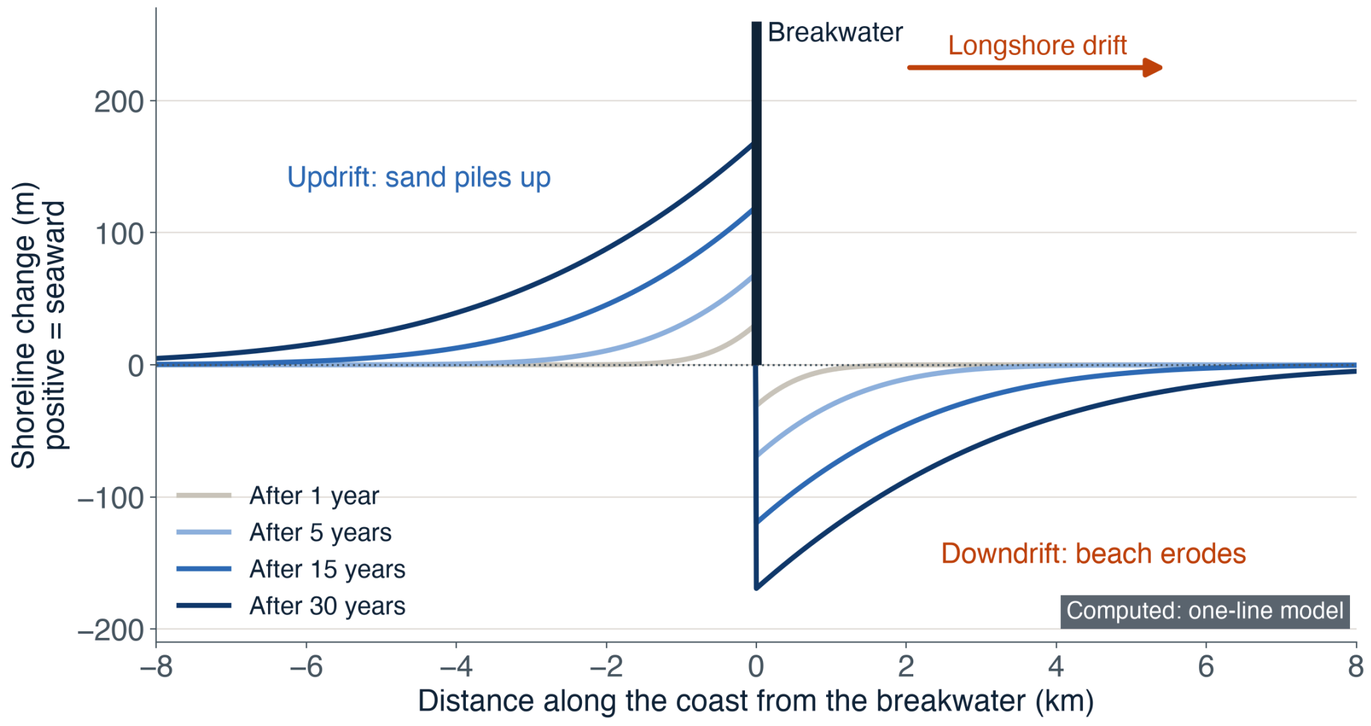


Year 18 **Updrift: +132 m (sand trapped)** **Downdrift: -132 m (beach eroded)**

Computed: one-line model (Pelnard-Considère). Not to scale across the shore.

CHANGE

The one-line model



Computed: Pelnard-Considère model, waves at about 3° to the shore.

$$\partial y/\partial t = \varepsilon\, \partial^2 y/\partial x^2$$

After 30 years: about 169 m gained beside the breakwater and the same lost on the other side.

The effect spreads with √t: about 6 km each way after 30 years.

Satellite shorelines test and calibrate such models.

CHANGE

Beyond the line: height and volume

Drones and lidar

Beach and dune height to a few centimetres: sand volume lost or gained, not just the shoreline.

InSAR on deltas

Sinking land behind the coast adds to relative sea-level rise (Lecture 6).

Satellite elevation

Stereo images and ICESat-2 laser profiles map coastal height; ICESat-2 also sees shallow seabed.

Why it matters

A beach can keep its shoreline while losing sand above it: volume tells you the buffer left before the next storm.

04

PART 4

Case studies

The world’s beaches, India’s coast, Puducherry and Satabhaya

CASE STUDY

The world’s sandy beaches

31%
Of the world’s ice-free coast is sandy

24%
Of sandy beaches eroding faster than 0.5 m/yr

28%
Accreting; the other 48% stable

1984–2016
33 years of Landsat images, analysed in the cloud

The first global map of beach change came entirely from free satellite images.

Erosion hotspots include protected areas and densely populated deltas.

Shows what Part 3’s methods can do at the scale of the planet.

Source: Luijendijk et al. (2018), ‘The State of the World’s Beaches’, Scientific Reports 8, 6641.

CASE STUDY

India’s coast, 1990–2018

33.6%
Of the mainland coast eroding

≈ 60%
Of West Bengal’s coast eroding

≈ 56%
Of Puducherry’s coast eroding

Atlas
Shoreline-change maps for every coastal state

Satellite images from 1990 to 2018, checked with field surveys.

Erosion hotspots near harbours, river mouths and sinking deltas.

Shoreline change also feeds the CRZ ‘hazard line’ mapped by the Survey of India.

Sources: NCCR (Ministry of Earth Sciences), National Assessment of Shoreline Changes along Indian Coast (2018); PIB; Down To Earth.

CASE STUDY

Puducherry: a harbour and a lost beach

1989
Harbour breakwaters completed

≈ 8 km
Of beach lost north of the harbour

2017–19
Restoration: submerged reef and sand nourishment

Jan 2019
Restored beach opened

Drift runs north: sand piled up south of the breakwaters, the town beach to the north disappeared.

A seawall of boulders replaced the beach for decades.

Satellite shorelines tracked both the loss and the recovery.

Sources: Frontiers in Marine Science (2024), hybrid beach restoration in Puducherry; Earth Journalism Network; NIOT.

CASE STUDY

Satabhaya, Odisha: villages lost to the sea

7 → 1
Villages: six taken by the Bay of Bengal

571
Families relocated in 2018

12 km
Inland to the Bagapatia resettlement colony

Kendrapara
District, on a low, eroding delta coast

Erosion and storm surges removed homes, fields and a temple.

Satellite shorelines document decades of retreat for planning and compensation.

Relocation solved the danger, not the loss of land and livelihoods.

Sources: Down To Earth; Eco-Business; CEEW.

05

PART 5

Engineering practice

Uses, coastal management, a workflow, limits and exercises

ENGINEERING

Where engineers use shoreline change

Setback and hazard lines

Keep new building back from the eroding shore; the CRZ hazard line includes shoreline change.

Harbour design

Predict and manage accretion and erosion either side.

Coastal protection

Decide where groynes, seawalls or nourishment are needed.

Checking projects

Did the nourishment or reef work? Track it year by year.

Infrastructure

Roads, rail, pipelines and outfalls near the shore.

Disaster response

Shoreline and dune loss after cyclones.

ENGINEERING

Managing an eroding coast

Hard

Seawalls, revetments, groynes, breakwaters. Protect what is behind, but often move erosion along the coast.

Soft

Beach nourishment, dune building, sand bypassing at harbours. Work with the drift, but need repeating.

Nature-based

Mangroves, dune vegetation, reefs. Trap sediment and absorb waves.

Retreat

Move people and structures back, as at Satabhaya. Sometimes the only lasting answer.

ENGINEERING

A shoreline-change workflow

01 Choose coast and years

02 Get images: Landsat, Sentinel-2

03 Extract waterlines (MNDWI)

04 Correct for tide

05 Baseline and transects

06 Rates: NSM, EPR, LRR

07 Check uncertainty

08 Map hotspots and act

Free tools: CoastSat (Python and Earth Engine), DSAS, QGIS.

Student project: rates for a beach near you, 1990 to today.

ENGINEERING

Limits and good practice

Limits

- Pixel size: metres, not centimetres
- Tides, waves and seasons shift the waterline
- Cloud during the monsoon (optical)
- A shoreline position says nothing about sand volume

Good practice

- Map one consistent indicator
- Correct to a tidal datum
- Use many images and regression rates
- Report uncertainty; call small rates ‘stable’

EXERCISES

Try these

01 An image was taken at a tide 0.8 m above mean sea level on a 1:25 beach. How far, and which way, is the mean-sea-level shoreline from the waterline?

02 On a transect the shoreline was 250 m from the baseline in 1990 and 180 m in 2025. Find the net movement and the end-point rate.

03 Each Landsat shoreline is uncertain by ±15 m. What is the uncertainty of a rate over 30 years? And Sentinel-2 (±5 m) over 10 years?

04 A house is 60 m from a shoreline eroding at 2.0 m/yr. When is it reached? What setback would a 50-year design life need?

EXERCISES

Answers

01 0.8 × 25 = **20 m**. At high tide the water is further up the beach, so the mean-sea-level shoreline is 20 m **seaward** of the waterline.

02 NSM = 180 − 250 = **−70 m** (landward); EPR = −70 ÷ 35 = **−2.0 m/yr**: erosion.

03 Landsat: $\sqrt{(15^2 + 15^2)} \div 30 \approx$ **0.71 m/yr**. Sentinel-2: $\sqrt{(5^2 + 5^2)} \div 10 \approx$ **0.71 m/yr**. Either way, rates under about ±0.7 m/yr are uncertain.

04 60 ÷ 2.0 = **30 years**. For 50 years: 2.0 × 50 = 100 m, so **more than 100 m** plus a buffer for storms and sea-level rise.

RECAP

Key points

01 Erosion is a sand-budget problem; structures and less river sand often drive it.

02 Water indices and sub-pixel edges find the waterline in satellite images.

03 Correct for tides, then measure change along transects from a baseline.

04 Use regression rates over long records, and respect the uncertainty.

05 Harbours trap sand on one side and starve the other: plan for both.

FURTHER READING

References

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Thank you. Questions?

Figures and the animation were made for this lecture. The breakwater figure and animation are computed from the Pelnard-Considère one-line model; the coastal scene, shorelines, transects and rates are simulated, and labelled as such; the beach profile is schematic. Case-study facts: Luijendijk et al. (2018); NCCR, PIB and Down To Earth (India); Frontiers in Marine Science and Earth Journalism Network (Puducherry); Down To Earth, Eco-Business and CEEW (Satabhaya). Sea-level rate from IPCC AR6. Full references on the previous slide.